Learning Python

★ For each problem, give yourself **30 minutes total**: think independently for the first **10 minutes**; if no ideas emerge, check the LeetCode topics and hints and think for **10 more minutes**; if you still have no direction after **20 minutes**, read the solution.

- `Counter(iterable)` (from `collections`) maps each element to its frequency and is the go-to tool for frequency counting in interview problems.
- `most_common(n)` returns the n most frequent `(element, count)` pairs in descending order (omit n for all), making it ideal for top-k frequency problems.
- `"".join([char for char in s if char.isalnum()])` strips a string to only letters and digits using `isalnum()`, which returns `True` for a–z, A–Z, and 0–9.
- `char.isnumeric()` accepts any Unicode numeric character (including fractions and superscripts), while `isdigit()` is stricter (ASCII digits 0–9 only).
- Never call `pop(i)` inside a loop that increments `i` forward, as each removal shifts remaining indices; a backward loop such as `for i in range(k, -1, -1): nums.pop(i)` is safe.
- `del nums[i]` is slightly faster than `nums.pop(i)` at the same O(n) complexity because `pop` has the extra overhead of returning the removed value, which can matter in tight loops over large lists.
- `list.index(i)` returns the index of the first occurrence of value `i` in the list, raising `ValueError` if it is not found.
- `list.count(value)` returns the number of times `value` appears in the list.
- `math.prod(list)` (from `math`) returns the product of all elements in the iterable, returning `1` for an empty iterable.
- `arr.insert(index, new_value)` inserts `new_value` at the given `index`, shifting all subsequent elements one position to the right.

- `my_dict.values()` returns a view object of all values in the dictionary, which can be saved into a list with `values_list = my_dict.values()`.
- `sum(my_dict.values())` returns the sum of all values in the dictionary, useful for quickly computing totals without an explicit loop.
- `max(my_dict.values())` returns the largest value in the dictionary, useful for finding the maximum frequency or count in a frequency map.
- `round(number, ndigits)` rounds `number` to `ndigits` decimal places (default 0), returning an `int` when called with no `ndigits` argument or a `float` otherwise.
- A **stack** in Python is simply a `list` where `append(val)` acts as push and `pop()` (no argument) acts as pop, both in O(1) time.
- Prefer `stack.append()` and `stack.pop()` over slice assignment such as `s = s[:-1]`, which creates a full O(n) copy each time and causes quadratic runtime in a loop.
- Two separate `if` statements in a loop are evaluated independently (both can trigger in the same iteration), whereas `if/elif` ensures only one branch fires per iteration.
- The **slow and fast pointer** technique (Floyd's cycle finding algorithm) uses two pointers where fast moves two steps per iteration and slow moves one; because their relative speed is one, they are guaranteed to meet if a cycle exists.
- To find the **nth node from the end** of a linked list, advance fast by n steps first, then move both slow and fast together one step at a time; when fast reaches the end, slow is pointing to the node whose `next` should be updated.

1 LeetCode Practice

Table 1: LeetCode practice tracker

Question	Level	Date	Revisit	Done
1. Arrays & Hashing				
217. Contains Duplicate	Easy			✓
1. Two Sum	Easy			✓
242. Valid Anagram	Easy			✓
383. Ransom Note	Easy			✓
349. Intersection of Two Arrays	Easy			✓
1512. Number of Good Pairs	Easy			✓
205. Isomorphic Strings	Easy			✓

Question	Level	Date	Revisit	Done
49. Group Anagrams	Medium			✓
347. Top K Frequent Elements	Medium			✓
238. Product of Array Except Self	Medium			✓
41. First Missing Positive	Hard	Nov 13		
2. Two Pointers				
344. Reverse String	Easy			✓
125. Valid Palindrome	Easy			✓
27. Remove Element	Easy			✓
26. Remove Duplicates from Sorted Array	Easy			✓
283. Move Zeroes	Easy			✓
977. Squares of a Sorted Array	Easy			✓
88. Merge Sorted Array	Easy			✓
167. Two Sum II: Input Array Is Sorted	Medium			✓
11. Container With Most Water	Medium			✓
15. 3Sum	Medium			✓
42. Trapping Rain Water	Hard	Nov 14		
3. Sliding Window				
643. Maximum Average Subarray I	Easy			✓
1876. Substrings of Size Three with Distinct Characters	Easy			✓
2269. Find the K-Beauty of a Number	Easy			✓
1984. Minimum Difference Between Highest and Lowest of K Scores	Easy			✓
1652. Defuse the Bomb	Easy			✓
219. Contains Duplicate II	Easy			✓
121. Best Time to Buy and Sell Stock	Easy			✓
3. Longest Substring Without Repeating Characters	Medium			✓
567. Permutation in String	Medium			✓
424. Longest Repeating Character Replacement	Medium			✓
76. Minimum Window Substring	Hard	Nov 17		
4. Stack				
20. Valid Parentheses	Easy			✓
682. Baseball Game	Easy			✓
1047. Remove All Adjacent Duplicates In String	Easy			✓
1544. Make The String Great	Easy			✓
844. Backspace String Compare	Easy			✓
232. Implement Queue using Stacks	Easy		⋆	
496. Next Greater Element I	Easy		⋆	
155. Min Stack	Medium			✓
150. Evaluate Reverse Polish Notation	Medium			✓
739. Daily Temperatures	Medium			✓
84. Largest Rectangle in Histogram	Hard	Nov 18		
5. Binary Search				
704. Binary Search	Easy			✓

Question	Level	Date	Revisit	Done
374. Guess Number Higher or Lower	Easy			✓
35. Search Insert Position	Easy			✓
278. First Bad Version	Easy			✓
744. Find Smallest Letter Greater Than Target	Easy			✓
367. Valid Perfect Square	Easy			✓
69. Sqrt(x)	Easy			✓
74. Search a 2D Matrix	Medium			✓
875. Koko Eating Bananas	Medium		⋆	
153. Find Minimum in Rotated Sorted Array	Medium			✓
4. Median of Two Sorted Arrays	Hard	Nov 19		
6. Linked List				
876. Middle of the Linked List	Easy		⋆	
83. Remove Duplicates from Sorted List	Easy			✓
203. Remove Linked List Elements	Easy			✓
206. Reverse Linked List	Easy		⋆	
141. Linked List Cycle	Easy	Nov 9	⋆	
160. Intersection of Two Linked Lists	Easy	Nov 9	⋆	
234. Palindrome Linked List	Easy	Nov 10		✓
19. Remove Nth Node From End of List	Medium	Nov 10	⋆	
2. Add Two Numbers	Medium	Nov 11	⋆	
143. Reorder List	Medium	Nov 11		
23. Merge k Sorted Lists	Hard	Nov 20		
7. Trees (DFS and BFS)				
94. Binary Tree Inorder Traversal	Easy	Nov 12		
104. Maximum Depth of Binary Tree	Easy	Nov 13		
226. Invert Binary Tree	Easy	Nov 14		
100. Same Tree	Easy	Nov 17		
101. Symmetric Tree	Easy	Nov 18		
112. Path Sum	Easy	Nov 19		
543. Diameter of Binary Tree	Easy	Nov 20		
102. Binary Tree Level Order Traversal	Medium	Nov 21		
98. Validate Binary Search Tree	Medium	Nov 24		
230. Kth Smallest Element in a BST	Medium	Nov 25		
8. Heap / Priority Queue				
1464. Maximum Product of Two Elements in an Array	Easy	Nov 26		
1046. Last Stone Weight	Easy	Nov 26		
2558. Take Gifts From the Richest Pile	Easy	Nov 27		
506. Relative Ranks	Easy	Nov 27		
1337. The K Weakest Rows in a Matrix	Easy	Nov 28		
2335. Minimum Amount of Time to Fill Cups	Easy	Nov 28		
703. Kth Largest Element in a Stream	Easy	Dec 1		
215. Kth Largest Element in an Array	Medium	Dec 1		
973. K Closest Points to Origin	Medium	Dec 2		
621. Task Scheduler	Medium	Dec 2		

Question	Level	Date	Revisit	Done
9. Backtracking				
509. Fibonacci Number	Easy	Dec 3		
231. Power of Two	Easy	Dec 3		
326. Power of Three	Easy	Dec 4		
590. N-ary Tree Postorder Traversal	Easy	Dec 4		
257. Binary Tree Paths	Easy	Dec 5		
1863. Sum of All Subset XOR Totals	Easy	Dec 5		
401. Binary Watch	Easy	Dec 8		
78. Subsets	Medium	Dec 8		
46. Permutations	Medium	Dec 9		
39. Combination Sum	Medium	Dec 9		
10. Graphs				
1791. Find Center of Star Graph	Easy	Dec 10		
997. Find the Town Judge	Easy	Dec 10		
1436. Destination City	Easy	Dec 11		
733. Flood Fill	Easy	Dec 11		
463. Island Perimeter	Easy	Dec 12		
1971. Find if Path Exists in Graph	Easy	Dec 12		
993. Cousins in Binary Tree	Easy	Dec 15		
200. Number of Islands	Medium	Dec 15		
994. Rotting Oranges	Medium	Dec 16		
207. Course Schedule	Medium	Dec 16		
11. Dynamic Programming				
70. Climbing Stairs	Easy	Dec 17		
1137. N-th Tribonacci Number	Easy	Dec 17		
746. Min Cost Climbing Stairs	Easy	Dec 18		
118. Pascal's Triangle	Easy	Dec 18		
119. Pascal's Triangle II	Easy	Dec 19		
338. Counting Bits	Easy	Dec 19		
392. Is Subsequence	Easy	Dec 22		
198. House Robber	Medium	Dec 22		
322. Coin Change	Medium	Dec 23		
300. Longest Increasing Subsequence	Medium	Dec 23		